Flexibility-Oriented Scheduling of Thermal and Pumped Hydro Units Using Demand Envelope Characterization Under Renewable Uncertainty

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***Abstract*—The increasing penetration of renewable energy sources introduces significant variability and uncertainty into power system operations, challenging the ability of conventional thermal units to maintain supply–demand balance. This paper proposes a flexibility-oriented scheduling framework that integrates demand envelope characterization with stochastic optimization of thermal generators and pumped storage hydro (PSH) units. First, a logistic demand envelope is extracted from five years of historical Great Britain load data to define upper and lower flexibility boundaries. Second, a scenario-based unit commitment and economic dispatch model is formulated, capturing generator ramp limits, startup costs, and PSH state-of-charge dynamics. Numerical results on a realistic system with 18 thermal units and 6 PSH units over a 12-hour horizon demonstrate that PSH significantly reduces flexibility shortages, but load shedding remains necessary under high-demand scenarios due to binding ramp constraints. The proposed framework provides grid operators with a practical tool for flexibility adequacy assessment and operational planning under renewable uncertainty.**

***Index Terms*—Flexibility envelope, pumped storage hydro, stochastic scheduling, unit commitment, renewable uncertainty.**

I. Introduction

A. Problem Statement

Modern power systems face unprecedented operational challenges due to the increasing penetration of variable renewable energy sources such as wind and solar power. Unlike conventional generation, renewable output is highly variable and partially unpredictable, resulting in rapid net-load fluctuations that require additional operational flexibility. Traditional unit commitment (UC) and economic dispatch (ED) formulations primarily focus on energy balance and generation adequacy, often neglecting the flexibility requirements associated with steep ramping events and renewable uncertainty.

As renewable penetration grows, insufficient operational flexibility may lead to renewable curtailment, reserve shortages, or controlled load shedding during peak ramp periods. Therefore, practical methods for quantifying and managing flexibility have become increasingly important for system operators.

B. Related Work

Several studies have investigated flexibility assessment and reserve management in modern power systems. Nosair and Bouffard [1] proposed energy-centric reserve definitions using flexibility envelopes derived from net-load trajectories. Li et al. [2] introduced a data-driven approach for flexibility characterization in distribution systems. Xu et al. [3] developed a two-layer generation expansion planning framework with flexibility balance constraints. Other important contributions include reserve quantification methods under wind integration [4], [5] and adaptive robust optimization approaches for flexibility aggregation [6].

Despite these advances, many existing approaches either focus on long-term planning, require complex data-driven models, or neglect the operational interaction between thermal generators and energy-limited storage resources.

C. Contributions

This paper proposes a flexibility-oriented scheduling framework with the following contributions:

1) Demand envelope extraction using logistic fitting of historical Great Britain load data to quantify flexibility requirements.
2) A stochastic UC/ED formulation incorporating thermal ramp-rate limits and PSH state-of-charge dynamics under multiple demand scenarios.
3) Quantitative assessment of flexibility shortages through load shedding analysis under renewable uncertainty.

The remainder of this paper is organized as follows. Section II presents the demand envelope characterization. Section III describes the stochastic scheduling formulation. Section IV discusses the case study and numerical results. Section V concludes the paper.

II. Demand Envelope Characterization

A. Logistic Demand Fitting

Historical load data from the Great Britain power system exhibit a characteristic intra-day pattern with morning and evening ramps. To capture the upper and lower flexibility boundaries, a logistic function is fitted to the aggregated demand trajectory:

$$f(x) = \frac{L}{1 + e^{-k(x-x_0)}} \tag{1}$$

where:

- L is the asymptotic maximum demand,
- k is the growth rate,

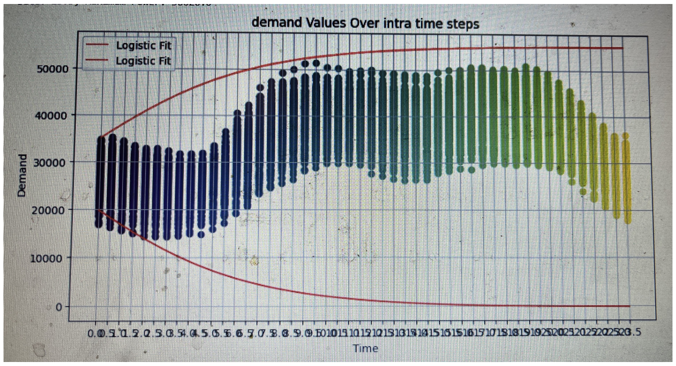


Fig. 1. Upper and lower GB demand envelopes obtained from logistic fitting of historical load data. The shaded region represents the required operational flexibility band.

- x_0 is the inflection point.

The upper and lower demand envelopes are defined as:

$$D^{upper}(t) = f(t) + \Delta(t) \tag{2}$$
$$D^{lower}(t) = -f(t) + C \tag{3}$$

where $\Delta(t)$ represents residual variability around the fitted curve.

Figure 1 illustrates the extracted demand envelope obtained from five years of historical Great Britain load data.

B. Flexibility Interpretation

The width of the envelope directly quantifies the ramping flexibility required from controllable generation resources. As the envelope widens during evening hours, additional operational flexibility becomes necessary to maintain reliability under uncertain renewable conditions.

III. Stochastic Scheduling Model

A. Objective Function

The scheduling problem minimizes the expected operational cost over a 12-hour horizon:

$$\min \sum_{\omega} \pi_\omega \left[\sum_{g\in G} (c_g P_{g,t,\omega} + SU_g u_{g,t,\omega}) + \sum_{s\in S} c_s P_{s,t,\omega} + VOLL \cdot LS_{t,\omega} \right] \tag{4}$$

where:

- $P_{g,t,\omega}$ is thermal generation output,
- $P_{s,t,\omega}$ is PSH generation output,
- $u_{g,t,\omega}$ is binary commitment status,
- $LS_{t,\omega}$ is load shedding,
- $VOLL$ is the value of lost load.

B. Constraints

1) Power Balance:

$$\sum_g P_{g,t,\omega} + \sum_s P_{s,t,\omega} + LS_{t,\omega} = D_{t,\omega} \tag{5}$$

2) Generator Limits:

$$P_g^{min} u_{g,t,\omega} \leq P_{g,t,\omega} \leq P_g^{max} u_{g,t,\omega} \tag{6}$$

3) Ramp Constraints:

$$P_{g,t,\omega} - P_{g,t-1,\omega} \leq RU_g \tag{7}$$
$$P_{g,t-1,\omega} - P_{g,t,\omega} \leq RD_g \tag{8}$$

4) PSH State of Charge:

$$SOC_{s,t} = SOC_{s,t-1} - \frac{P_{s,t}}{\eta_d}\Delta t \tag{9}$$

subject to:

$$0 \leq SOC_{s,t} \leq SOC_s^{max} \tag{10}$$
$$0 \leq P_{s,t} \leq P_s^{rated} \tag{11}$$

C. Demand Scenarios

Three demand scenarios are constructed from the extracted envelope:

- Low demand scenario,
- Medium demand scenario,
- High demand scenario.

Table I summarizes representative demand levels.

IV. Case Study and Results

A. Test System

The test system consists of 18 thermal generating units with a combined capacity of approximately 6200 MW and 6 PSH units with a total capacity of 900 MW.

Thermal generators have ramp rates between 50–80 MW/h. Each PSH unit is rated at 150 MW with:

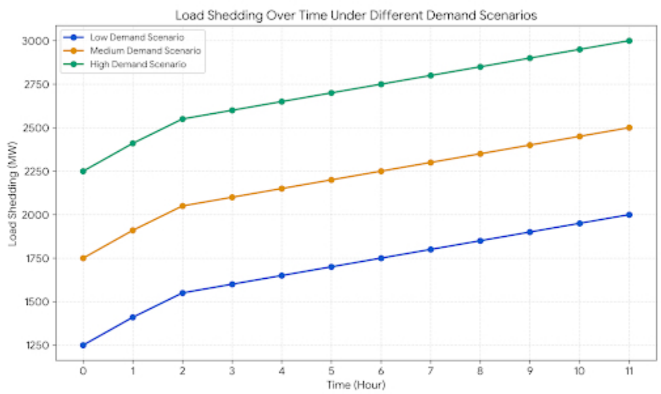
- discharge efficiency: 0.9,
- initial SOC: 1200 MWh,
- maximum SOC: 1500 MWh.

The optimization problem is formulated as a mixed-integer linear program and solved using PuLP with the GLPK solver.

Table II summarizes representative generator parameters.

TABLE I
Demand Scenarios Derived from GB Demand Envelope

Time (h)	Low (MW)	Medium (MW)	High (MW)
0	4200	4600	5000
3	4500	4900	5300
6	5000	5500	6000
9	5600	6100	6700
11	5800	6400	7000



B. Optimal Dispatch and Load Shedding

The optimal expected operating cost is 8.42×10^8 nominal units.

Thermal generation follows the demand trajectory and approaches maximum output during evening peak periods. Several units simultaneously hit ramp-rate constraints during steep load transitions.

PSH units provide fast-response flexibility support and reduce net-load variability. However, due to state-of-charge limitations, their flexibility contribution gradually decreases over time.

Table III summarizes load shedding values under different demand scenarios.

TABLE II
SELECTED THERMAL GENERATOR AND PSH PARAMETERS

Unit	**P$_{min}$ (MW)**	**P$_{max}$ (MW)**	**Ramp Rate (MW/h)**
gen05	100	320	70
gen11	120	350	80
gen53	100	300	65
PSH (each)	0	150	150

Figure **??** illustrates the load shedding trajectories for the three demand scenarios.

TABLE III
LOAD SHEDDING AT SELECTED HOURS

Time (h)	**Low (MW)**	**Medium (MW)**	**High (MW)**
0	120	250	420
6	180	350	600
11	250	500	850

C. Comparison With and Without PSH

To evaluate the flexibility contribution of pumped storage hydro, the scheduling problem was solved both with and without PSH resources.

The results confirm that PSH significantly improves operational flexibility and reduces both operating cost and flexibility shortages.

D. Flexibility Insights

Insight 1: PSH substantially mitigates flexibility shortages but cannot completely eliminate load shedding under severe ramping conditions.

Insight 2: Thermal generator ramp-rate limits become binding during rapid evening load increases.

Insight 3: The widening demand envelope accurately identifies periods of elevated operational stress and flexibility scarcity.

V. CONCLUSION

This paper presented a flexibility-oriented scheduling framework combining demand envelope characterization with stochastic optimization of thermal and pumped hydro units under renewable uncertainty.

The proposed demand envelope approach provides a practical and computationally efficient method for quantifying flexibility requirements. Numerical results demonstrate that PSH significantly improves operational flexibility and reduces load shedding. However, flexibility shortages may still occur under severe ramping conditions due to thermal ramp-rate constraints and storage energy limitations.

Future work will incorporate charging dynamics of PSH units, demand response participation, and rolling-horizon stochastic scheduling.

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REFERENCES

[1] H. Nosair and F. Bouffard, “Energy-centric flexibility management in power systems,” *IEEE Transactions on Power Systems*, vol. 31, no. 6, pp. 5071–5081, 2016.

[2] Q. Li, J. Liu, B. Cui, W. Song, and J. Ye, “Distribution system flexibility characterization: A network-informed data-driven approach,” *IEEE Transactions on Power Systems*, 2023.

[3] J. Xu et al., “Two-layer generation expansion planning based on flexibility balance,” *IEEE Access*, vol. 11, pp. 23456–23470, 2023.

[4] Y. Makarov, C. Loutan, and P. de Mello, “Operational impacts of wind generation on California power systems,” *IEEE Transactions on Power Systems*, vol. 24, no. 2, pp. 1039–1050, 2009.

[5] R. Doherty and M. O’Malley, “A new approach to quantify reserve demand in systems with significant installed wind capacity,” *IEEE Transactions on Power Systems*, vol. 20, no. 2, pp. 587–595, 2005.

[6] X. Chen and N. Li, “Leveraging two-stage adaptive robust optimization for power flexibility aggregation,” *IEEE Transactions on Smart Grid*, vol. 12, no. 5, pp. 3954–3965, 2021.